SUPER BOWL ECONOMICS & GAME DYNAMICS

WHY STUDY THE SUPER BOWL?

01.

ONE OF THE MOST WATCHED EVENTS IN THE U.S.

02.

COMPANIES SPEND MILLIONS ON ADS — HAVE GROWN SIGNIFICANTLY OVER THE YEARS

03.

EXPLORING:

- ADVERTISING TRENDS
- GAME COMPETITIVENESS

RESEARCH QUESTION:

HOW HAVE SUPER BOWL ADVERTISING COSTS CHANGED OVER TIME, AND ARE GAME CHARACTERISTICS SUCH AS SCORING AND COMPETITIVENESS RELATED TO ONE ANOTHER?

DATA:

CONTENT

- Source: Kaggle – Super Bowl Game Results dataset
- Time span: 1967–2018 (52 games)
- Data collected from:
- NFL records
- TV networks & ratings agencies

kaggle

KEY VARIABLES

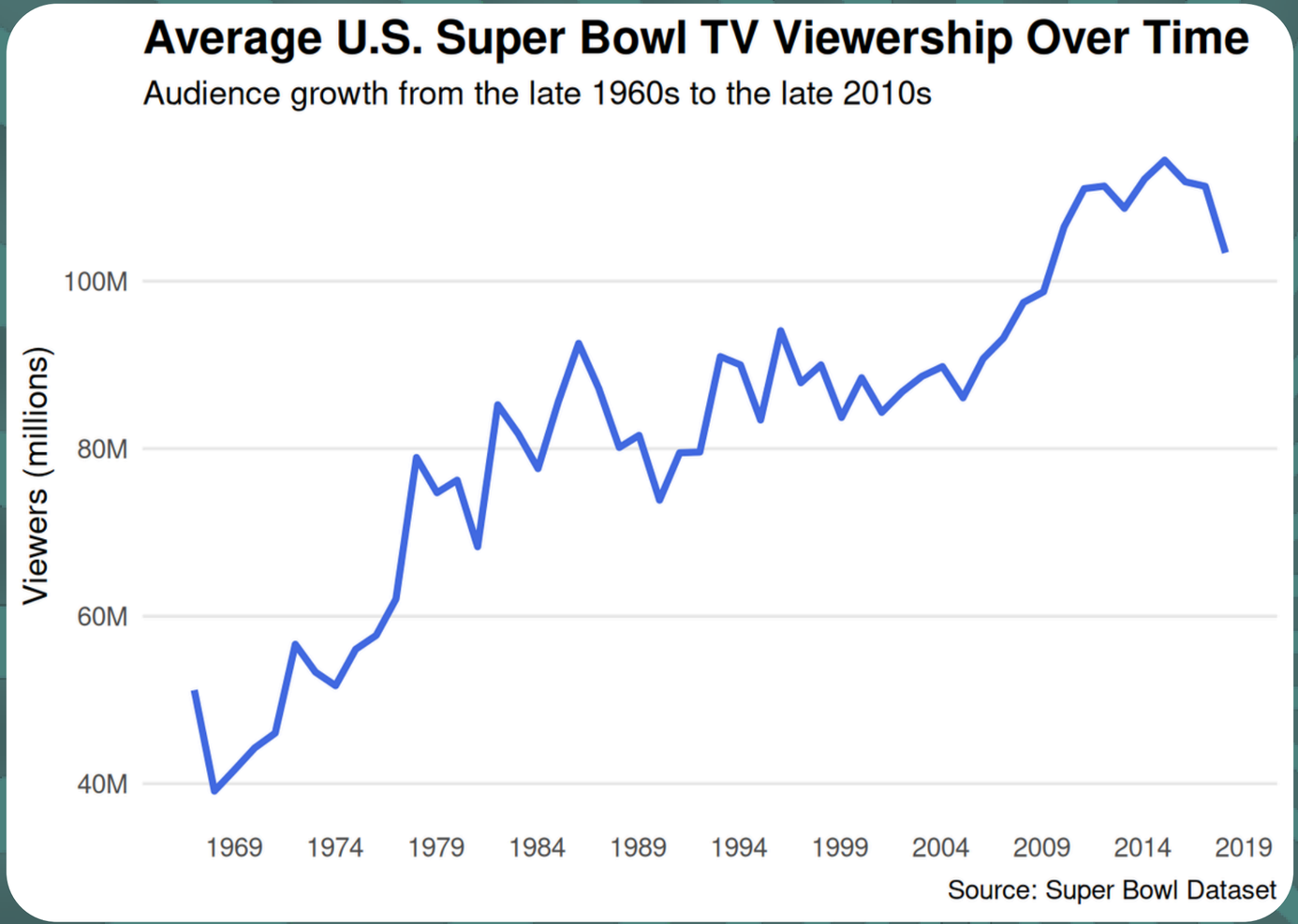
- Cost per 30s ad
- Average U.S. viewers
- Network
- Winning & losing team points

DERIVED VARIABLES:

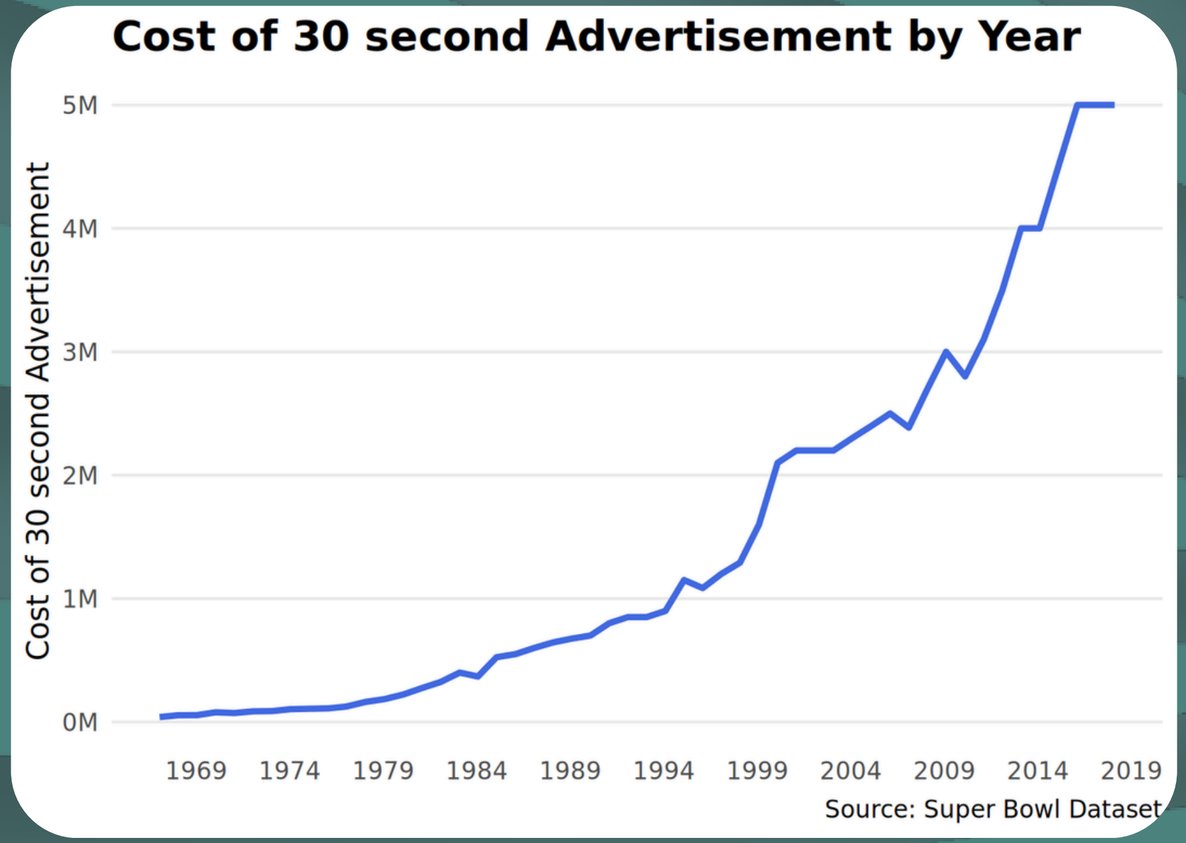
- Year
- Margin of victory
- Total points

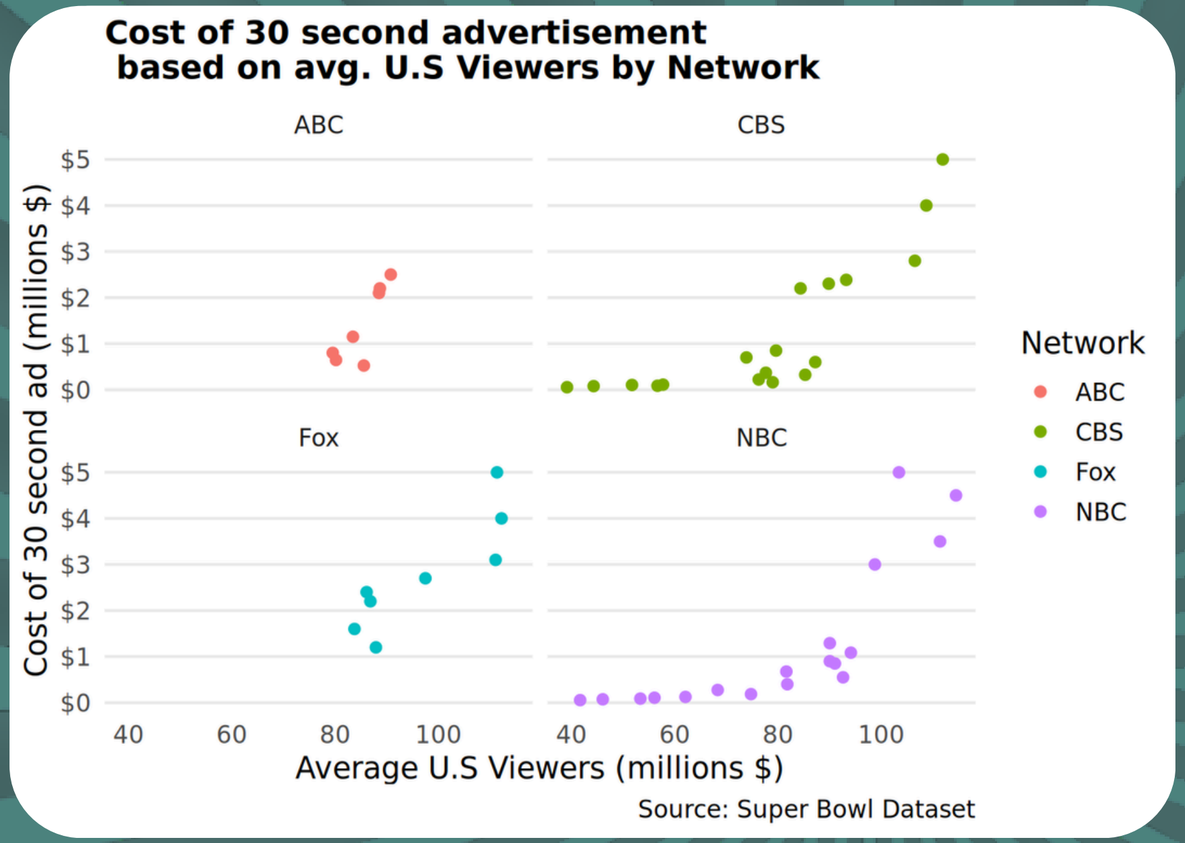
VIEWERSHIP OVER TIME

- VIEWERSHIP INCREASES FORM ~40 MIL TO OVER 100 MIL
- AUDIENCE GROWTH REFLECTS INCREASING POPULARITY OF THE SUPER BOWL



AD COST OVER TIME BY NETWORK





- STRONG UPWARD TREND IN AD COSTS OVER TIME
- VERY STRONG LINEAR RELATIONSHIP (R = 0.931)
- ~86.6% OF VARIATION EXPLAINED BY YEAR
- ~$92K INCREASE PER YEAR ON AVERAGE

DISCUSSION:

KEY FINDINGS:

- STRONG POSITIVE RELATIONSHIP BETWEEN YEAR AND AD COST
- INCREASES WITH HIGHER VIEWERSHIP
- CONSISTENT ACROSS BROADCASTING NETWORKS

INTERPRETATION:

- REFLECTS LONG-TERM GROWTH IN THE COMMERCIAL VALUE OF MASS-VIEWERSHIP EVENTS
- COMPANIES INCREASINGLY WILLING TO INVEST FOR LARGE AUDIENCE REACH

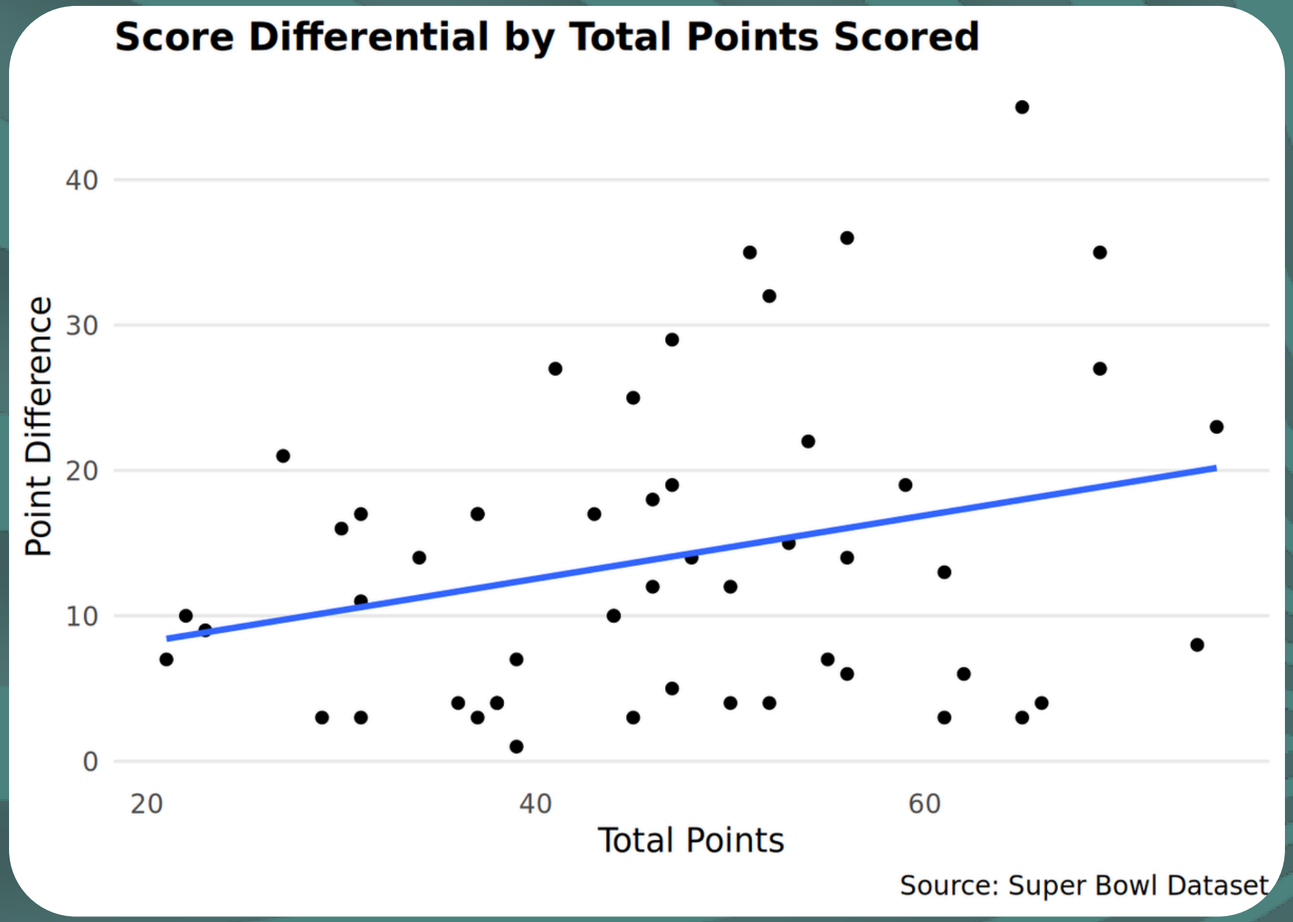
LIMITATIONS:

- OBSERVATIONAL DATA → NO CAUSATION
- OTHER VARIABLES (I.E. INFLATION)

FUTURE WORK:

- INCORPORATE ADDITIONAL MARKET VARIABLES (ROI, ENGAGEMENT)

TOTAL POINTS & MARGIN OF VICTORY



- NO CLEAR PATTERN BETWEEN TOTAL POINTS AND POINT DIFFERENCE
- WEAK LINEAR RELATIONSHIP (R = 0.280)
- R^2 = 0.078 → ONLY ~8% OF VARIATION EXPLAINED
- HIGH-SCORING GAMES ARE NOT NECESSARILY CLOSE GAMES

DISCUSSION:

KEY FINDINGS:

- WEAK RELATIONSHIP BETWEEN MARGIN OF VICTORY AND TOTAL POINTS
- COMPETITIVENESS IS NOT A STRONG PREDICTOR OF SCORING

INTERPRETATION:

- SCORING INFLUENCED BY:
  - OFFENSE & DEFENSE STRENGTH
  - GAME STRATEGY
- CLOSE GAMES ≠ ALWAYS HIGH SCORING

LIMITATIONS:

- SMALL SAMPLE SIZE
- SIMPLE LINEAR MODEL

FUTURE WORK:

- INCLUDE MORE VARIABLES
- USE MORE ADVANCED MODELS

THANK YOU!